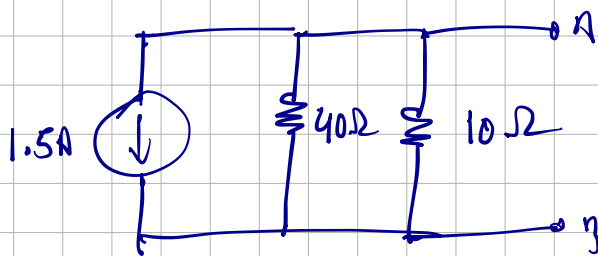
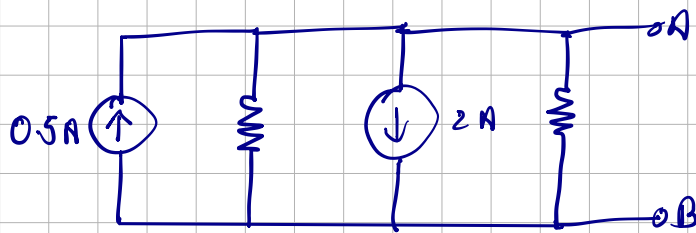
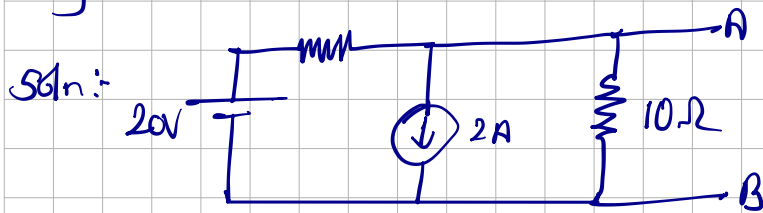


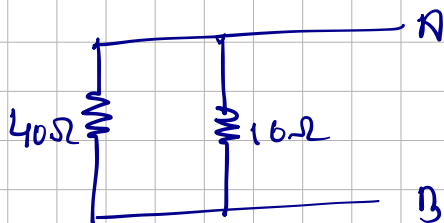
NAME :- SHAHU ROHIT KAMLESH

Roll No :- 25f3100145

Q.1]



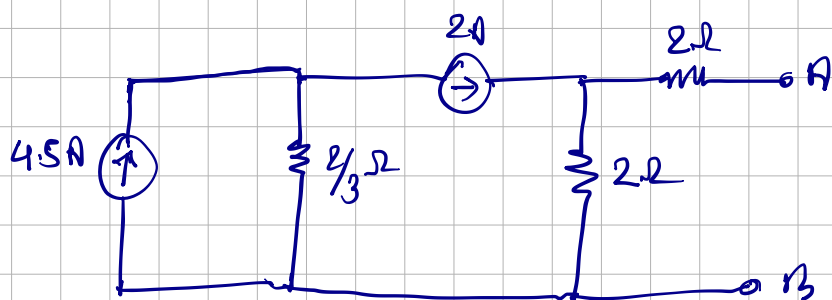
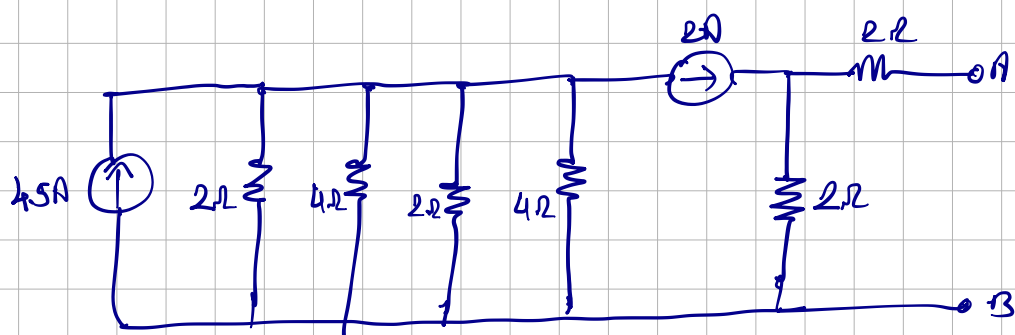
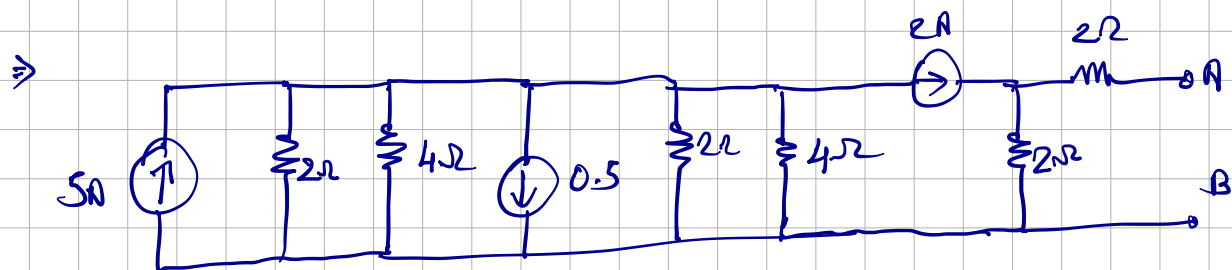
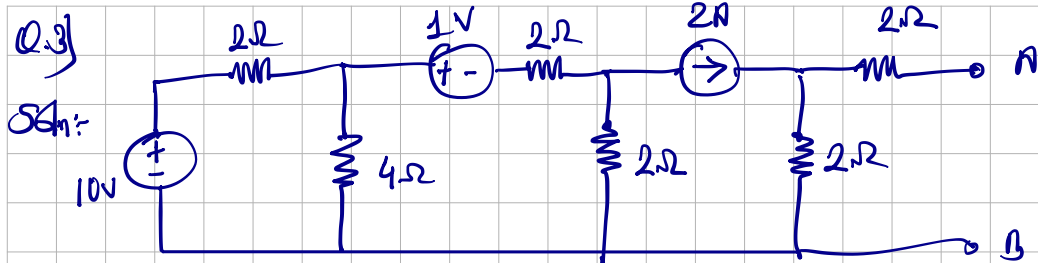
Q.2]



$$R_{AB} = \frac{10 \times 40}{10 + 40}$$

$$= \frac{10 \times 40}{50}$$

$$= 8\Omega$$



now we have to find voltage across AB

$$\therefore V_{AB} = 2A \times 2\Omega$$

$$V_{AB} = 4V$$

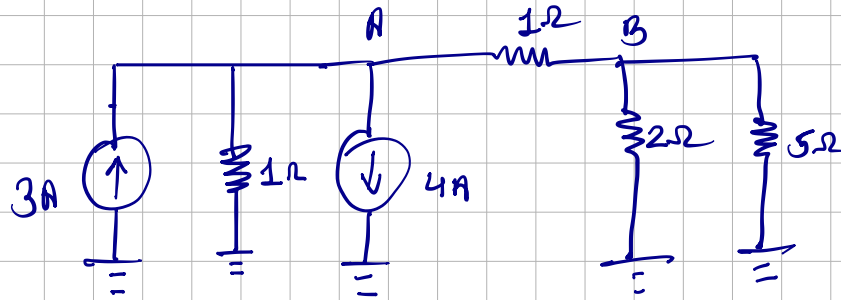
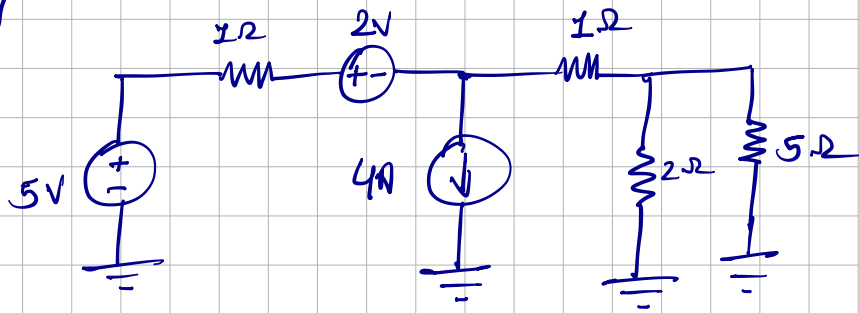
$$\therefore V_{TH} = 4V$$

Q.4]

$$R = (2 + 2)\Omega$$

$$= 4\Omega$$

Q.5)



So, it's clear by nodal analysis at node A
by KCL

The current flowing through $B \rightarrow A$ is $1A$

Since 2Ω & 5Ω are in parallel

$\therefore$ using current division rule.

$$\therefore \frac{2 \times 1}{7} = \frac{2}{7} A.$$

$$\therefore \text{Power} = I^2 R$$

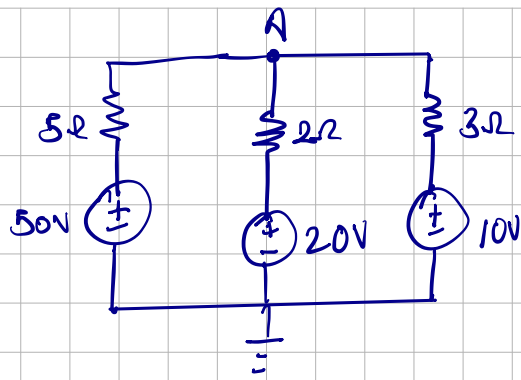
$$= \left(\frac{2}{7}\right)^2 \times 5\Omega$$

$$= \frac{4}{49} \times 5 W$$

$$= \frac{20}{49} W = 0.408 W$$

Q.6)

Soln:-



"Using K.C.I at Node A"

$$\frac{V_A - 50}{5} + \frac{V_A - 20}{2} + \frac{V_A - 10}{3} = 0$$

$$\frac{V_A}{5} - 10 + \frac{V_A}{2} - 10 + \frac{V_A}{3} - \frac{10}{3} = 0$$

$$\frac{V_A}{5} + \frac{V_A}{2} + \frac{V_A}{3} = 20 + \frac{10}{3}$$

$$\frac{V_A}{5} + \frac{5V_A}{6} = \frac{60+10}{3}$$

$$\frac{31V_A}{30} = \frac{70}{3}$$

$$V_A = \frac{70 \times 10}{31}$$

$$V_A = 23.34 \text{ V}$$

∴ Current through 50V source

$$= \frac{50 - 23.34}{5}$$

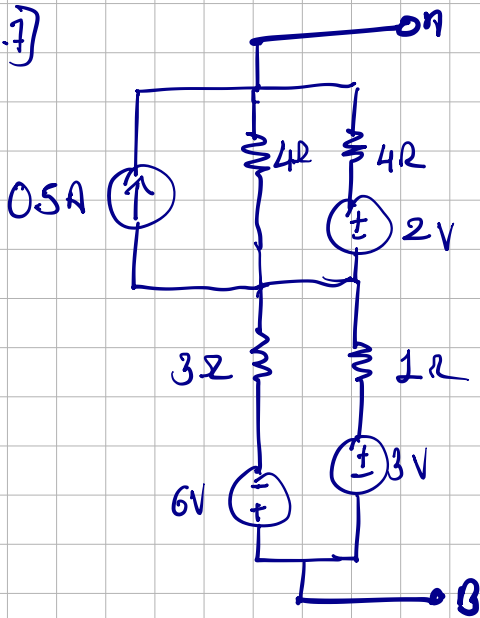
$$= 5.332 \text{ A}$$

$$\therefore \text{power delivered} = -V \cdot I$$

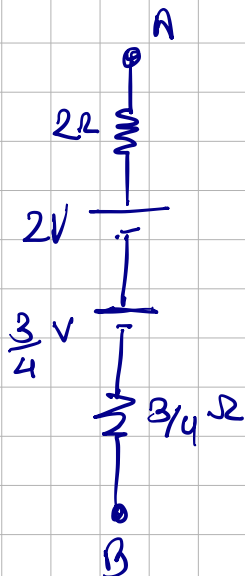
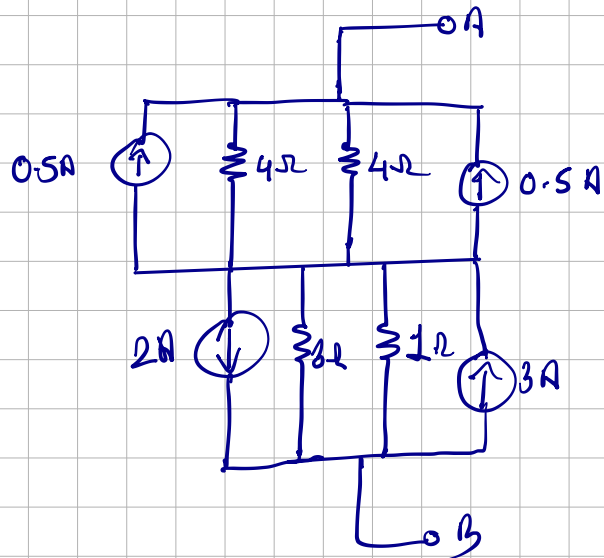
$$= -80 \times 5.332 \text{ W}$$

$$= -266.6$$

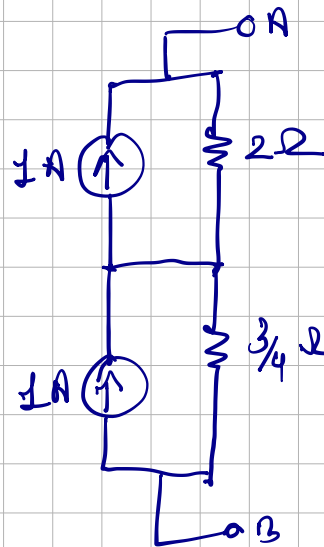
Q.7]



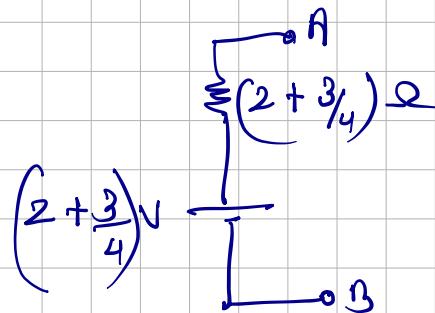
$\Rightarrow$



$\leftarrow$



$\rightarrow$



$$\therefore V_{TH} = 2 + 0.75 = 2.75 \text{ V}$$

Q.8]

$$\Rightarrow R_{TH} = \left(2 + \frac{3}{4}\right) \Omega$$
$$= 2.75 \Omega$$

Q.9]

Soln:-

$$V(t) = 10 \sin(\omega t + 30^\circ)$$

$$i(t) = 10 \sin(\omega t - 30^\circ)$$

$\therefore$ Angle between voltage & current = 60°

$$\therefore \text{power consumed} = V_{rms} \cdot I_{rms} \cos \phi$$

$$= \frac{10}{\sqrt{2}} \cdot \frac{10}{\sqrt{2}} \cdot \frac{1}{2}$$

$$= \frac{5}{2}$$

$$= 25 \text{ W}$$

Q.10]

$$\text{Soln:- } \cos \phi = \cos 45^\circ = \frac{1}{\sqrt{2}}$$

$$\therefore \text{Apparent power} = V_{rms} \cdot I_{rms}$$

$$= 1000 \text{ W}$$